Graphic Era HILL UNIVERSITY

End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: B. Tech. C.E and 7th Semester
Name of the Paper: Design of Steel Structures
Paper Code: TCE-701
Time: 3-hour

Maximum Marks: 100

Note:
(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question are 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)

a. Explain the mechanical properties of structural steel relevant to design. (CO1)

b. Illustrate the Limit State Design method as per IS 800:2007 with an example. (CO1)

c. Find the efficiency of the following riveted joints: (CO1)

Double riveted lap joint of 6 mm plates with 20 mm diameter rivets having a pitch of 65 mm.

Permissible tensile stress in plate = 120 MPa

Permissible shearing stress in rivets = 90 MPa

Permissible crushing stress in rivets = 180 MPa

Q2. (2X10=20 Marks)

a. Describe the steps involved in designing a tension splice. (CO2)

b. Compare the advantages and disadvantages of using Tee sections versus angle sections in tension members. (CO2)

c. Write down the Procedure for the design of Tension Members. (CO2)

Q3. (2X10=20 Marks)

a. What are the design Steps of Compression Members? (CO3)

b. Discuss the advantages and limitations of lacing compared to battening in column design. (CO3)

c. What is a Lug Angle and what are its requirements as per IS 800: 2007? (CO3)

Q4. (2X10=20 Marks)

a. Derive an expression for the Flexural Strength of Laterally Supported Beam. (CO4)

b. Describe the procedure for designing a built-up beam with welded connections. (CO4)

c. Explain the importance of lateral-torsional buckling in beam design. (CO4)

Q5. (2X10=20 Marks)

a. What are the different types of Roof trusses? (CO5)

b. Explain the various terms used in Roof Trusses. (CO5)

c. Explain the procedure for calculating the design loads on roof trusses as per IS codes. (CO5)